

Written Report – 6.419x Module 1

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▪ Problem 1.1

1. (2 points) How would you run a randomized controlled double-blind experiment to determine the effectiveness of the vaccine? Write down procedures for the experimenter to follow. (Maximum 200 words)

Solution: From a large sample of participants, randomly assign to either the treatment group or the control group. This random assignment is crucial to ensure both groups are statistically similar, minimizing the impact of selection bias.

Stratification of group of people into different demographics like age, gender and regions and then selecting at random from these individual groups reduces the effect of confounding variable.

The experiment should be double-blind, meaning neither the participants nor the nurse administering the injections or collecting the data know who is receiving the vaccine and who is receiving the placebo. This prevents bias from influencing the results. A third party, with no direct involvement in the study's execution, should manage the randomization and coding of the injections.

2. (3 points) For each of the NFIP study, and the Randomized controlled double blind experiment above, which numbers (or estimates) show the effectiveness of the vaccine? Describe whether the estimates suggest the vaccine is effective.

Solution:

Although, the polio rate in the vaccinated group (25) is significantly lower than in the unvaccinated group (54). The NFIP study does have good control variables to clearly say its effectiveness among all demographics. The unvaccinated group (grade 1 and grade3) and non-consent grade 2 are not good control variables to begin with against grade 2 vaccinated children.

Randomized Controlled Double-Blind Experiment measures effectiveness by comparing the number of polio cases in the treatment group (vaccinated) to the number of polio cases in the control group (salt injection). The vaccinated group had a much lower number of polio cases (**28**) compared to the placebo group (**71**). This large difference, free from the biases of the observational study, provides strong evidence that the vaccine works to prevent polio.

(3-1). (a) (2 points) Scenario: What if Grade 1 and Grade 3 students are different from Grade 2 students in some ways? For example, what if children of different ages are susceptible to polio in different degrees? Can such a difference influence the result from the NFIP experiment? If so, give an example of how a difference between the groups can influence the result. Describe an experimental design that will prevent this difference between groups from making the estimate not reliable.

Solution: Yes, differences between the grade levels can influence the result of the NFIP study, making the estimate unreliable.

Example: Suppose older (Grade 3) children are naturally less susceptible to polio than younger children (Grade 1). If a disproportionate number of unvaccinated children were from Grade 3, while most vaccinated children were from Grade 1, the higher polio rate in the unvaccinated group could be partly or entirely due to this age difference, not the lack of vaccination. This is a form of confounding.

Instead of comparing among grade1 or grade 2 children, one should select at random from these two groups and segregate them into treatment and control group.

(3-2). (2 points) Polio is an infectious disease. The NFIP study was not done blind; that is, the children know whether they get the vaccine or not. Could this bias the results? If so, Give an example of how it could bias the results. Describe an aspect of an experimental design that prevent this kind of bias.

Solution: Yes, the lack of blinding in the NFIP study could bias the results. This is a type of selection bias and response bias.

Example: Parents of vaccinated children might be less vigilant about their children's health, assuming they are protected. Conversely, parents of unvaccinated children might be more likely to keep them away from sick children or report any suspicious symptoms to doctors, who might then be more inclined to diagnose even mild cases as polio. This differential reporting could inflate the number of polio cases in the unvaccinated group, making the vaccine appear more effective than it truly is.

A double-blind experimental design prevents this kind of bias. In a double-blind study, neither the participants nor the researchers know who is receiving the vaccine and who is receiving a placebo. This ensures that:

- Participants' behavior is not influenced by their vaccination status.
- The doctors and researchers collecting the data are not influenced by their knowledge of which group the participants belong to.

(3-3). (2 points) Even if the act of "getting vaccine" does lead to reduced infection, it does not necessarily mean that it is the vaccine itself that leads to this result. Give an example of how this could be the case. Describe an aspect of experimental design that would eliminate biases not due to the vaccine itself.

Solution: One scenario where a reduced infection rate isn't solely due to the vaccine is when act of getting vaccine changes behavior or have other psychological effects. It's not the vaccine itself that's preventing the infection, but these changes in behaviors.

To eliminate this type of bias, a placebo group should be included in the experimental design. Instead of simply not getting a vaccine, the control group would receive a salt injections. This ensures that any behavioral changes or psychological effects associated with the act of getting an injection and visiting the clinic are present in both groups, isolating the true effect of the vaccine itself.

4. (2 points) In both experiments, neither control groups nor the no-consent groups got the vaccine. Yet the no-consent groups had a lower rate of polio compared to the control group. Why could that be?

Solution: The non-consent groups had a lower rate of polio due to inherent bias of non-consent groups. Parents who were more educated and had higher incomes were more likely to consent to their children being part of the experiment. These families tended to live in more sanitary and less crowded environments. Conversely, the parents who did not consent to the trial were, on average, from lower-income and less hygienic backgrounds. Their children were more likely to have been exposed to the virus at a young age, often developing a mild, non-paralytic form of the disease and building natural immunity.

5.(3 points) In the randomized controlled trial, the children whose parents refused to participate in the trial got polio at the rate of 46 per 100000, while the children whose parents consented to participate got polio at a slighter higher rate of 49 per 100000 (treatment and control groups taken together). On the basis of these numbers, in the following year, some parents refused to allow their children to participate in the experiment and be exposed to this higher risk of polio. Were their conclusion correct? What would be the consequence if a large group of parents act this way in the next year's trial?

Solution: No, the parents' conclusion that their children were at a higher risk of polio by participating in the trial was incorrect. This is because it didn't account for the selection bias present in the study population. The group of children whose parents consented to the study and the group who didn't were fundamentally different. The "consenters" group, whose parents were often more educated and from a higher socioeconomic status, lived in cleaner, more hygienic environments. This meant they had less prior exposure to the poliovirus and thus had no natural immunity, making them more susceptible to the paralytic disease if exposed.

The slightly higher polio rate reflected this pre-existing difference in natural immunity between the two populations.

If a large group of parents were to act this way in the next year's trial, the integrity and reliability of the experiment would be compromised. The randomized controlled trial's success relies on a large and representative sample size to ensure that the treatment and control groups are as similar as possible.

▪ Problem 1.3

(a-1)(2 points) Your colleague on education studies really cares about what can improve the education outcome in early childhood. He thinks the ideal planning should be to include as much variables as possible and regress children's educational outcome on the set. Then we select the variables that are shown to be statistically significant and inform the policy makers. Is this approach likely to produce the intended good policies?

Solution: No, colleague's approach is unlikely to produce good policies. Basing policy decisions solely on statistical significance from a regression model with many variables is problematic and can lead to flawed conclusions. This method, sometimes called "p-hacking" or "data dredging," involves selectively reporting statistically significant results, which can produce a spurious excess of positive findings. A small p-value does not necessarily imply a large or important effect. It can be achieved with a large sample size even if the effect is tiny and has no practical or educational significance. Good policies should be informed by many contextual factors, not just p-values, including the quality of measurements, the study's design, and external evidence.

(a-2). (3 points) Your friend hears your point, and think it makes sense. He also hears about that with more data, relations are less likely to be observed just by chance, and inference becomes more accurate. He asks, if he gets more and more data, will the procedure he proposes find the true effects? Hint: You might need to design some experiment.

Solution: While it's true that with more data, chance findings become less likely, your friend's proposed procedure will still have a major flaw and may not find the "true" effects. The ASA statement highlights that with a large enough sample size, any effect, no matter how small or unimportant, can produce a small p-value. This is a critical point. A statistically significant result ($p < 0.05$) simply indicates a low probability of observing the data if the null hypothesis were true, not that the null hypothesis is false or that the effect is meaningful.

To demonstrate this, consider a thought experiment:

- Start with the null hypothesis: Imagine there is truly no effect of 100 different educational policies on a child's outcome.
- Generate a small dataset: With a small dataset and a significance threshold of $p=0.05$, you would expect about 5 of these 100 policies to show a "statistically significant" effect purely by random chance.
- Generate a huge dataset: Now, repeat the experiment with a massive dataset. Due to the large sample size, a tiny, practically meaningless, and truly non-existent effect will almost certainly be flagged as statistically significant. The p-values for all 100 policies will likely become very small, but you won't be able to distinguish the policies with a true effect from those with a tiny, meaningless effect.

Therefore, simply getting more and more data will only exacerbate the problem. It will make more effects appear "statistically significant" even when they are not scientifically or practically important.

(b-2). (2 points) A neuroscience lab is interested in how consumption of sugar and coco may effect development of intelligence and brain growth. They collect data on chocolate consumption and number of Nobel prize laureates in each nation, and finds the correlation to be statistically significant. Should they conclude that there exists a relationship between chocolate consumption and intelligence?

Solution: Our previous discussion highlights the dangers of relying on observational studies and p-values alone. While an observational study might find a correlation between people who eat chocolate and higher intelligence scores, this does not prove that chocolate *causes* the improved scores. Many other factors could be at play (confounding variables), such as socioeconomic status, diet, or lifestyle. People with higher incomes might be able to afford more chocolate and have better access to education and nutrition, which are the true causes of their higher intelligence.

(b-3). (1 point) In order to study the relation between chocolate consumption and intelligence, what can they do?

Solution: To study the relationship between chocolate consumption and intelligence, researchers should perform randomized controlled trial (RCT). This experimental design directly tests for a causal relationship, avoiding the pitfalls of "p-hacking" and spurious correlations that can arise from observational studies.

- **Recruit Participants:** Find a large group of people and measure their baseline intelligence scores using a standardized test.
- **Random Assignment:** Randomly divide the participants into two groups: **Group A (Treatment):** This group receives a specific dose of chocolate (e.g., a certain amount of dark chocolate with a known flavonoid content) to consume daily for a set period. **Group B (Control):** This group

receives a placebo (e.g., a similar-looking white chocolate with no flavonoids) or no chocolate at all.

- **Blinding:** To avoid bias, the participants and the researchers measuring the outcome should not know which group is the treatment group.
- **Measure Outcome:** After the study period, both groups are given the intelligence test again.
- **Analyze the Results:** Researchers would then compare the change in intelligence scores between the two groups. If the group that consumed chocolate shows a statistically significant improvement compared to the control group, this provides strong evidence that chocolate consumption *caused* the increase in intelligence.

***(b-4). (3 points)** The lab runs a randomized experiment on 100 mice, add chocolate in half of the mice's diet and add in another food of the equivalent calories in another half's diet. They find that the difference between the two groups time in solving a maze puzzle has p-value lower than 0.05. Should they conclude that chocolate consumption leads to improved cognitive power in mice?*

Solution: No, they should conclude that chocolate consumption leads to improved cognitive power.

A p-value doesn't indicate the probability that the hypothesis is true. It only measures the incompatibility of the data with the null hypothesis (that there is no difference between the two groups of mice). It does not tell you if chocolate caused the effect.

Scientific conclusions should not be based solely on a p-value threshold. Relying on a "bright-line" rule like $p < 0.05$ oversimplifies the interpretation of data. The researchers need to consider other factors, such as the actual size of the effect (how much faster did the chocolate-fed mice solve the maze?), the precision of their measurements, and other context-specific information.

P-values do not measure the importance of a result. A small p-value can be obtained even with a very tiny, practically meaningless effect, especially with a large sample size. In this case, while 100 mice is a decent sample, a small p-value on its own doesn't tell you if the difference in maze-solving time is significant enough to matter in a real-world context.

***(b-5). (3 points)** The lab collects individual level data on 50000 humans on about 100 features including IQ and chocolate consumption. They find that the relation between chocolate consumption and IQ has a p-value higher than 0.05. However, they find that there are some other variables in the data set that has p-value lower than 0.05, namely, their father's income and number of siblings. So they decide to not write about chocolate consumption, but rather, report these statistically significant results in their paper, and provide possible explanations Is this approach correct?*

Solution:No, this approach is not correct and is a classic example of p-hacking or selective reporting, which the ASA statement explicitly warns against. The lab's initial goal was to study the relationship between chocolate consumption and IQ. When they found a non-significant result, they did not report it. Instead, they "fished" for other statistically significant findings from the many variables they collected.

This violates Principle 4 of the ASA statement, which states: "Proper inference requires full reporting and transparency. Researchers should avoid p-hacking—selectively reporting analyses with certain p-values—which renders the reported p-values uninterpretable."

By testing 100 different features, the researchers have an extremely high chance of finding a few that are statistically significant by random chance alone, even if no true relationship exists. At a 0.05 significance level, you would expect 5 out of 100 variables to appear significant by chance. Reporting only these few significant findings misrepresents the scientific evidence and can lead to false conclusions.

(c). (3 points) A lab just finishes a randomized controlled trial on 10000 participants for a new drug, and find a treatment effect with p-value smaller than 0.05. After a journalist interviewed the lab, he wrote a news article titled "New trial shows strong effect of drug X on curing disease Y." Is this title appropriate? What about "New drug proves over 95% success rate of drug X on curing disease Y"?

Solution: Neither title is appropriate, as they both draw conclusions that are not supported by the p-value alone.

"New trial shows strong effect of drug X on curing disease Y."

This title is misleading because a p-value less than 0.05 does not imply a "strong effect." The p-value only indicates that the observed result is statistically significant—unlikely to have occurred by random chance if the null hypothesis were true. As mentioned in the ASA statement, a statistically significant result can be produced by a very small, practically insignificant effect if the sample size is large enough. A trial with 10,000 participants could find a tiny effect to be statistically significant. A journalist should instead report the actual effect size and its confidence interval to give readers a true sense of the drug's impact.

"New drug proves over 95% success rate of drug X on curing disease Y."

This title is also incorrect and dangerous. The most common misinterpretation of a p-value is equating $p=0.05$ with a 95% chance that the hypothesis is true. This is a fundamental misunderstanding of what a p-value represents. As Principle 2 of the ASA statement clarifies, a p-value does not measure the probability that a hypothesis is true. It measures the incompatibility of the data with a specified statistical model (the null hypothesis). The phrase "over 95% success rate" is a statement about a probability of success, which cannot be derived from a p-value.

(d). (1 point) Your boss wants to decide on company's spending next year. He thinks letting each committee debates and propose the budget is too subjective a process and the company should learn from its past and let the fact talk. He gives you the data on expenditure in different sectors and the company's revenue for the past 25 years. You run a regression of the revenue on the spending on HR sector, and find a large effect, but the effect is not statistically significant. Your boss saw the result and says "Oh, then we shouldn't increase our spending on HR then".

Is his reasoning right?

Solution: No, boss's reasoning is not right. A non-significant p-value does not mean that there is no effect. The large, but not statistically significant, effect suggests that the data is not precise enough to rule out the possibility of a strong relationship.

Instead of focusing on the p-value, boss should look at the **effect size** and its **confidence interval**. The large effect you found suggests that increasing HR spending could plausibly lead to a large increase in revenue. The confidence interval would show a range of plausible values for that effect. If the range includes a substantial positive effect, it would be unwise to conclude that there is no relationship and to cut HR spending.

(e). (1 point) Even if a test is shown as significant by replication of the same experiment, we still cannot make a scientific claim.

True or False?

Solution: False. The replication of a finding through repeated, well-designed experiments is a cornerstone of the scientific method. When an effect is consistently observed across multiple, independent replications of the same experiment—especially by different research teams—it strengthens the evidence for a scientific claim. While a single, statistically significant finding can be due to random chance (the p-value is a probability, after all), its persistence across replications suggests the effect is real and not a fluke.

(f). (2 points) Your lab mate is writing up his paper. He says if he reports all the tests and hypothesis he has done, the results will be too long, so he wants to report only the statistical significant ones.

Is this OK? If not, why?

Solution:

No, this is not okay. The lab mate's approach is a form of selective reporting, also known as "p-hacking" or "data dredging," which is a serious ethical and methodological breach.

The American Statistical Association's statement on p-values is very clear on this point. Principle 4 states that "Proper inference requires full reporting and transparency" and that selectively reporting statistically significant results "renders the reported p-values uninterpretable." When you run multiple tests and only report the ones that "worked" (i.e., had a p-value below 0.05), you are essentially guaranteeing that you will find some "significant" results by random chance. This inflates the false positive rate and creates a misleading impression of the evidence.

(g). (2 points)

If I see a significant p-values, it could be the case that the null hypothesis is consistent with truth, but my statistical model does not match reality.

True or False?

Solution: True. A significant p-value indicates that your data is unlikely given the null hypothesis, but it does not guarantee that the null hypothesis is false. The calculation of the p-value relies on a statistical model, which includes assumptions about the data's distribution and the relationships between variables.

As Principle 1 of the ASA statement notes, "A p-value indicates the incompatibility of the data with a specified statistical model." If your statistical model is a poor representation of reality, the p-value may be small not because the null hypothesis is false, but because the model's assumptions are incorrect. For example, if you incorrectly assume a linear relationship when the true relationship is non-linear, a p-value may show a significant effect even if the null hypothesis is consistent with reality. Good statistical practice requires careful consideration of whether the statistical model is appropriate for the data and the research question.

▪ **Problem 1.5**

(8) (3 points) Show that the extent of repeated independent testing by different teams can reduce the probability of the research being true.

Solution:

To show how repeated testing can reduce the probability of a research finding being true, we can use the formula for the Positive Predictive Value (PPV) that was previously introduced. PPV=

$$\frac{\text{P}(\text{relation exists, at least one of the } n \text{ repetitions finds significant})}{\text{P}(\text{at least one of the } n \text{ repetitions finds significant})} = \frac{\text{P}(\text{at least one significant} \mid \text{relation}) \times \text{P}(\text{relation})}{\text{P}(\text{at least one significant})}$$

$$\text{P}(\text{at least one significant}) = \text{P}(\text{at least one significant} \mid \text{relation}) \times \text{P}(\text{relation}) + \text{P}(\text{at least one significant} \mid \text{no relation}) \times \text{P}(\text{no relation})$$

$$\text{However, } \text{P}(\text{at least one significant} \mid \text{relation}) = 1 - \text{P}(\text{none of the } n \text{ teams finds a significant result} \mid \text{relation}) = (1 - \beta^n)$$

$$\text{Similarly, } \text{P}(\text{at least one significant} \mid \text{no relation}) = 1 - \text{P}(\text{none of the } n \text{ teams finds a significant result} \mid \text{no relation}) = (1 - (1 - \alpha)^n)$$

$$\text{P}(\text{at least one significant}): (1 - \beta^n) \frac{R}{R+1} + (1 - (1 - \alpha)^n) \frac{1}{R+1}$$

And therefore
$$PPV = \frac{R}{R + \frac{1 - [1 - \alpha]^n}{1 - \beta^n}}$$

The core of the argument comes from examining the denominator. As the number of teams, n , increases, the term $(1 - (1 - \alpha)^n)$ **becomes** very close to 1. This means that as more teams test the same hypothesis, the probability of at least one of them getting a false positive approaches certainty. At the same time, $(1 - \beta^n)$ also approaches 1 but at a slower rate because $1 - \alpha > \beta$ (Type 2 error). Therefore, the overall PPV decreases.

(9). (2 points) What would make bias or increasing teams testing the same hypothesis not decrease PPV? (Hint: Please treat the two issues separately.)

Solution: Bias or selective reporting would not decrease the PPV if **all results were reported, regardless of statistical significance**. If every study, including those with null results, were published, a single false-positive finding from one team would be seen in the context of all the other non-significant results. Increasing team testing would not decrease the PPV if the ratio of the number of “true relationships” to “no relationships” among those tested in a scientific field (R) are **extremely high**.

The PPV formula is:

$$PPV = \frac{R}{R + \frac{1 - [1 - \alpha]^n}{1 - \beta^n}}$$

However, if R is very large, the ratio approaches 1:

$$PPV \approx \frac{R}{R} \approx 1$$

And therefore, do not depend on n .

(10). (5 points) *Read critically and critique! Remember the golden rule of science, replication? For the third table in the paper, if researchers work on the same hypothesis but only one team finds significance, the other teams are likely to think the results are not robust, since it is not replicable. In light of this, how would you model the situation when multiple teams work on the same hypothesis and the scientific community requires unanimous replication? What would be the PPV? (You do not need to include a bias term for this question.)*

Solution: For a hypothesis to be considered “true” by the scientific community, all n independent teams must find a significant result.

In this case, there is a unanimous significant result from all n teams.

The PPV formula is:

$$\frac{P(\text{relations exists, all } n \text{ repetitions find significant})}{P(\text{all } n \text{ repetitions find significant})} = \frac{P(\text{all significant} | \text{true}) P(\text{true})}{P(\text{all significant} | \text{true}) P(\text{true}) + P(\text{all significant} | \text{false}) P(\text{false})}$$

$$\text{Now, } P(\text{all significant} | \text{true}) P(\text{true}) = (1 - \beta)^n \frac{R}{R + 1}$$

Also, $P(\text{all significant} \mid \text{true}) P(\text{true}) + P(\text{all significant} \mid \text{false}) P(\text{false}) = (1 - \beta)^n \frac{R}{R+1} + \alpha^n \frac{1}{R+1}$

Therefore, $PPV = \frac{R}{R + \frac{\alpha^n}{(1-\beta)^n}}$ and as n increases, PPV increases because $\alpha < 1-\beta$ for $\alpha=0.05$

(11). (3 points) Suppose there is no bias and no teams are racing for the same test, so there is no misconduct and poor practices. Will publications still be more likely to be false than true?

Solution: Yes, publications can still be more likely to be false than true because of the inherent statistical realities of research. For a single, well-conducted study, the PPV is calculated as:

$$PPV = \frac{R(1-\beta)}{R(1-\beta) + \alpha}$$

In this model, where: R is the ratio of the number of "true relationships" to "no relationships" among those tested in a scientific field. $(1-\beta)$ is the statistical power of the study. α is the significance level (0.05).

A finding is more likely to be false than true when the PPV is less than 50% ($PPV < 0.50$). This occurs when the denominator is more than twice the numerator, which simplifies to the condition: $R(1-\beta) < \alpha$. Even with no misconduct, this condition can be met in many research settings, especially in new or exploratory fields. For instance, if a field has a low pre-study odds (low R) and/or studies with low statistical power ($1-\beta$), the likelihood of a false positive (a Type I error) can easily outweigh the likelihood of a true positive.

(12). (2 points) In light of this paper, let's theoretically model the problem mentioned by the American Statistical Association about p-values concern! Suppose people base the decision to making scientific claim on p-values, which parameter does this influence? R , α or β ? Describe the effect on the PPV if scientists probe random relations and just look at p-value as a certificate for making scientific conclusion.

Solution: This problem, where people rely on p-values to make scientific claims, primarily influences two parameters in our model: R and α .

Effect on R (Ratio of True to False Relationships)

When scientists "probe random relations," it means they are testing a vast number of hypotheses without a strong theoretical basis or prior evidence. In this scenario, the denominator of R (the number of no-relationships) becomes much larger than the numerator (the number of true relationships).

Effect on α (Type I Error Rate)

In an ideal world with no bias, this would be a fixed parameter. However, the ASA statement highlights that this can lead to poor practices like p-hacking or selective reporting, which inflate the number of false positives. This effectively makes the real, or "effective," α much higher than the nominal 0.05.

Effect on PPV

The combined effect will decrease the PPV value significantly.

